

A choice experiment for testing stochastic discrete choice conditions

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Abstract

We run an experiment covering 32 different choice domains, including fine art, cars and
pizza. For each domain, there is a set of 5 related objects, and 26 menus—the subsets of two
or more objects. We measure population-level choice probabilities by presenting each menu to
40 participants and observing their choices of a single object. The purpose of the experiment
is to allow tests of various conditions over choice probabilities, over a wide variety of choice
domains. It is particularly well suited for testing conditions, like regularity, random utility and
the multiplicative inequality, which constrain choice probabilities across all menus of a domain.
Each participant faces exactly one menu from each domain. Participants are representative of
the Canadian adult population. For each domain and each of the three conditions above, we
report Bayes factors giving evidence for or against the condition. Evidence supports regularity
and random utility, conditions that most discrete models satisfy, across all domains; for the
multiplicative inequality, a condition satisfied by independent random utility models and the
Elimination by Aspects model, evidence ranges from favorable to strongly unfavorable.

Key words: Discrete choice, Stochastic choice, Choice experiment, Context effects, Random
utility, Bayesian inference

JEL codes: C11, C12, C15, C35, C91, D01, D90.

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1 Introduction

Discrete choice models, which describe the choices of decision makers from finite menus of options, are used extensively in economics, psychology, marketing, transportation research, health research and other disciplines. Most empirical applications seek to measure how attributes of choice options affect choice probabilities in particular cases; for example, how commute times and other features affect the probabilities of choosing various transportation options, in Train (1978). Here, we contribute experimental data to a distinct and more abstract research agenda that collects evidence for or against, as the case may be, various testable conditions on choice probabilities, without regard to object attributes, across a wide variety of choice environments.

Consider the following choice experiment, one of 32 we performed in the course of an internet survey. We asked each participant “Which one of the following cities do you think is furthest north?” and then presented them with one *menu*, a list of at least two options taken from the following set of five cities: Warsaw, Poland; London, United Kingdom; Vancouver, Canada; Paris, France; and Seattle, United States. The experimental manipulation consisted of varying the menu across participants. There are 26 possible menus from the set of five cities and we recruited 1040 participants. We randomly partitioned the participants into 26 groups of 40, and presented a different menu to each group. Table 1 shows the choice counts we observed, by menu and option.

The following examples illustrate several themes of the paper. From the menu of Vancouver and Paris, 17 participants chose Vancouver and 23 chose Paris; from the menu of Vancouver, Paris and Seattle, 26 chose Vancouver; 14, Paris; and none, Seattle. It appears that adding Seattle to the menu of Vancouver and Paris increases the probability that participants choose Vancouver. This would be a violation of a condition on choice probabilities known as *regularity*, which holds that if you add an option to a menu, the probability of choosing an existing option cannot increase.

Regularity is one of many stochastic choice conditions that have been studied in the choice literature. We will see later that it is a condition that holds for a large class of choice models. It is a hypothesis that can be tested directly using data, allowing us to draw conclusions about entire classes of models without needing to estimate model parameters. It jointly constrains choice probabilities across all 26 menus of the five-object universe described above. It is an abstract condition: it makes no reference to attributes of various options and can therefore be applied across very different choice environments. Finally, it is easy to understand and has some intuitive appeal as a plausible description of choice.

Menu A	Choice counts					Choice proportions				
	$N_A(a)$	$N_A(b)$	$N_A(c)$	$N_A(d)$	$N_A(e)$	$\hat{P}_A(a)$	$\hat{P}_A(b)$	$\hat{P}_A(c)$	$\hat{P}_A(d)$	$\hat{P}_A(e)$
$\{a, b\}$	22	18	-	-	-	0.550	0.450	-	-	-
$\{a, c\}$	31	-	9	-	-	0.775	-	0.225	-	-
$\{b, c\}$	-	29	11	-	-	-	0.725	0.275	-	-
$\{a, b, c\}$	17	16	7	-	-	0.425	0.400	0.175	-	-
$\{a, d\}$	25	-	-	16	-	0.610	-	-	0.390	-
$\{b, d\}$	-	32	-	8	-	-	0.800	-	0.200	-
$\{a, b, d\}$	24	14	-	2	-	0.600	0.350	-	0.050	-
$\{c, d\}$	-	-	17	23	-	-	-	0.425	0.575	-
$\{a, c, d\}$	30	-	6	4	-	0.750	-	0.150	0.100	-
$\{b, c, d\}$	-	24	14	2	-	-	0.600	0.350	0.050	-
$\{a, b, c, d\}$	23	4	12	1	-	0.575	0.100	0.300	0.025	-
$\{a, e\}$	30	-	-	-	10	0.750	-	-	-	0.250
$\{b, e\}$	-	31	-	-	9	-	0.775	-	-	0.225
$\{a, b, e\}$	19	15	-	-	6	0.475	0.375	-	-	0.150
$\{c, e\}$	-	-	36	-	4	-	-	0.900	-	0.100
$\{a, c, e\}$	30	-	9	-	2	0.732	-	0.220	-	0.049
$\{b, c, e\}$	-	22	15	-	3	-	0.550	0.375	-	0.075
$\{a, b, c, e\}$	9	15	14	-	2	0.225	0.375	0.350	-	0.050
$\{d, e\}$	-	-	-	26	14	-	-	-	0.650	0.350
$\{a, d, e\}$	26	-	-	5	9	0.650	-	-	0.125	0.225
$\{b, d, e\}$	-	25	-	8	7	-	0.625	-	0.200	0.175
$\{a, b, d, e\}$	16	18	-	3	3	0.400	0.450	-	0.075	0.075
$\{c, d, e\}$	-	-	26	14	0	-	-	0.650	0.350	0.000
$\{a, c, d, e\}$	33	-	4	1	2	0.825	-	0.100	0.025	0.050
$\{b, c, d, e\}$	-	24	13	3	0	-	0.600	0.325	0.075	0.000
$\{a, b, c, d, e\}$	19	9	10	2	0	0.475	0.225	0.250	0.050	0.000

Table 1: Observed choice counts and proportions for the “Latitude” domain, where the prompt is ‘Which one of the following cities do you think is furthest north?’ and the universe of objects is $\{a, b, c, d, e\}$, where a is Warsaw, Poland; b is London, United Kingdom, c is Vancouver, Canada; d is Paris, France; and e is Seattle, United States. Each row gives counts and proportions for the menu shown in the first column. The second through sixth columns show the number of participants who chose the objects a , b , c , d and e , respectively, from the menu. Where an object is not in the menu, a dash is displayed. The final five columns show the corresponding choice proportions. Analogous tables for all domains are in the Supplementary Materials.

Regularity, and other conditions implied by various classes of models, are targets of a large literature on context effects. This literature gathers empirical evidence against these conditions, and by extension, the models that imply them. The attraction effect, for example, occurs when adding an option that is clearly inferior to one option in a menu increases the probability of choosing the superior option, a clear violation of regularity. We designed the Latitude domain to try to elicit an attraction effect, based on our conjecture that it would be obvious to most participants, drawn from a panel of Canadian residents, that Vancouver is north of Seattle and would be seen as a clearly superior option. Indeed, from the menu consisting of these two cities, 37 chose Vancouver and 3 chose Seattle. The choice counts reported above reveal that when Seattle is added to the menu of Vancouver and Paris, the proportion of participants choosing Vancouver increased. This is suggestive evidence for the attraction effect, but to assess the strength of the evidence, we must analyse sampling variation.

The main contribution of this paper is a publicly available dataset designed for testing abstract stochastic choice conditions, such as regularity, across a wide range of choice domains. The experiment includes 32 diverse choice domains, including fine art, cars, and pizza. Some domains are specifically designed to elicit the attraction effect and other context effects. We do this to stress-test conditions on choice probabilities that are inconsistent with the effects and, more generally, to elicit patterns that many consider anomalous. Measuring choice probabilities on all menus allows us to jointly test the complete set of implications of a condition, such as regularity, that jointly constrains choice probabilities over all menus.

Section 2 develops the probabilistic framework. It documents three conditions on choice probabilities in detail, including regularity, and points to others in the literature. It describes three context effects and outlines the motivation for designing some of the domains to elicit them. Section 3 describes the experimental design in detail. Section 4 reviews the Bayesian models and inferential methods used in the empirical illustrations. These methods are adapted from earlier work but are summarized here to make the paper reasonably self-contained and reproducible. Section 5 illustrates some analyses that are possible using our data, including tests of the three conditions described in Section 2 and an example of testing for context effects. Section 6 concludes. All data and code are available in the Supplementary Materials.

2 Framework

Here we set up the probabilistic choice framework by defining a *random choice model*. We then define three conditions on choice probabilities we investigate in a later empirical section: regularity, the Block-Marschak inequalities and the multiplicative inequality. We then describe versions of three context effects that motivated parts of our experimental design: the attraction, similarity and compromise effects. Finally, we discuss individual- and population-level choice experiments and identify an aggregation property relating conditions on individual choice to conditions on population choice.

2.1 Random choice models

A random choice model specifies choice probabilities for a set $U = \{a_1, a_2, \dots, a_n\}$ of related choice objects, which we will call a universe. For each of the 32 choice domains in our experiment, there is a universe containing five related and easily comparable choice objects. We will write $U = \{a, b, c, d, e\}$ for a generic five-element universe. For the “Latitude” domain of the experiment, where participants see the prompt question “Which one of the following cities do you think is furthest north?”, a is Warsaw, b is London, c is Vancouver, d is Paris and e is Seattle.

Each participant sees a single menu from each domain and must choose a single option from each menu. Different participants see different menus and thus we need to define choice probabilities for each menu and each option within it. For a given universe U , let A be any non-empty subset of U and x be any option in A . Define $P_A(x)$ as the probability of choosing x from A . One of the 26 menus is $\{a, b\}$; $P_{\{a,b\}}(a)$ and $P_{\{a,b\}}(b)$ are the probabilities of choosing Warsaw and London from the menu consisting of these two cities. We learn something about these probabilities from the experiment: 22 participants chose Warsaw and 18 chose London, as we can see in the first row of Table 1, which tabulates choice counts for each of the 26 menus of the “Latitude” domain. We also define, for observed choice counts, $\hat{P}_A(x)$ as the proportion of times that an object x is chosen from a menu A , for $x \in A \subseteq U$. For example, we compute the observed choice proportions $\hat{P}_{\{a,b\}}(a) = 0.55$ and $\hat{P}_{\{a,b\}}(b) = 0.45$ for the “Latitude” domain. Although we defined menus earlier as choice sets with at least two elements, the notation $P_A(x)$ allows for choice sets A with a single element. This is to ensure that we can concisely write down conditions like the Block-Marschak inequalities below; the choice probabilities $P_{\{a\}}(a)$, for $a \in U$, are trivially equal to one.

A random choice model for the universe U is the complete specification of all choice probabilities

$P_A(x)$ for all $x \in A \subseteq U$, and is denoted P . In the case of repeated trials, we assume that observed choices are statistically independent, both within and across menus. We also assume that for each menu, observed choices are identically distributed. Thus a random choice model P for a universe gives a complete probabilistic description of an observed sequence of choices from menus of that universe. In a manner analogous to the definition of P , we define $\hat{P}$ as the complete specification of all the choice proportions $\hat{P}_A(x)$ observed in an experiment.

The term random choice model is often used to describe logit and other, sometimes parametric, sets of random choice models. We will refer to these as *classes* of random choice models.

One particularly important class of random choice models consists of those that describe the behaviour of random utility maximizers. Each time such a decision maker encounters a menu in a choice trial, they draw random utility values for all objects in the universe and then choose the object in that menu with the highest utility. Random utility is drawn independently of the prevailing menu. From trial to trial, draws are independent and identically distributed. Within a trial, however, the utilities of the various objects may be statistically dependent. We will call the condition defining this class of models “random utility”, for brevity. Most classes of random choice models used in applied work satisfy random utility, including the Logit and Probit classes. For a detailed exposition of random utility and how it relates to random choice models, using the same notation as this paper, see McCausland et al. (2020).

2.2 Global testable conditions for stochastic choice

Our term “condition” covers various axioms, constraints, hypotheses, properties, propositions and rules about probabilistic choice behaviour that can be expressed as restrictions over the various choice probabilities of a random choice model.

We have already seen regularity, the condition that if you add options to a menu, the probability of choosing an element already there does not go up. Formally, a random choice model P for a universe U of choice objects satisfies regularity if and only if for all $x \in A \subseteq B \subseteq U$,

$$P_A(x) \geq P_B(x).$$

Luce & Suppes (1965) describe regularity and show that it is necessary for random utility.

We have seen that regularity constrains choice probabilities across all menus. In fact, we can say something stronger. Every $P_A(\cdot)$ is *pivotal*, in the sense that for any random choice model

satisfying regularity and any menu A , one can change $P_A(\cdot)$ so that the modified model violates regularity. This is a compelling reason to collect choice data from every menu of a universe, if one wants to test regularity for some domain.

The following condition is relevant because it is not only necessary but also sufficient for random utility. It is far less interpretable than regularity, although see Barberá & Pattanaik (1986) for an interpretation of the sum in (1) for given A and x .

A random choice model P on a universe U satisfies the Block-Marschak inequalities if and only if for all non-empty $A \subseteq U$ and all $x \in A$,

$$\sum_{B: A \subseteq B \subseteq U} (-1)^{|B \setminus A|} P_B(x) \geq 0, \quad (1)$$

where $|B \setminus A|$ is the number of elements in B that are not also in A . ($\setminus$ is the set difference operator and $|\cdot|$ denotes cardinality.) Block & Marschak (1960) show that the Block-Marschak inequalities are necessary for random utility and Falmagne (1978) shows they are sufficient. For the Block-Marschak inequalities, as for regularity, each $P_A(x)$ is pivotal: since regularity is necessary for random utility, the same change in $P_A(\cdot)$ that breaks regularity breaks the Block-Marschak inequalities.

Sattath & Tversky (1976) introduce the multiplicative inequality, claiming “it has some appeal as a principle of individual choice behaviour”. It states that the probability of choosing an object x from a union $A \cup B$ of two menus is at least as great as the probability of choosing x from A and then again from B in independent trials. Suppose, for example, a random choice model P on the universe of the “Latitude” domain satisfies the multiplicative inequality. Then the probability of choosing Warsaw from the menu of Warsaw, London and Vancouver must be at least as great as the joint probability of two independent events: choosing Warsaw over London in pairwise choice and then choosing Warsaw again in pairwise choice, this time over Vancouver. In the experiment, 17 out of 40 participants chose Warsaw from the tripleton menu, 22 out of 40 chose Warsaw over London and 31 out of 40 chose Warsaw over Vancouver. Since $0.425 = \hat{P}_{\{a,b,c\}}(a) < \hat{P}_{\{a,b\}}(a)\hat{P}_{\{a,c\}}(a) = 0.42625$, the system of choice *proportions* $\hat{P}$ violates the multiplicative inequality. A formal test of the multiplicative inequality for choice *probabilities* will take into account sampling variation and the full set of inequalities.

Formally, a random choice model P satisfies the *multiplicative inequality* if and only if for all

$$x \in A, B \subseteq U,$$

$$P_{A \cup B}(x) \geq P_A(x)P_B(x).$$

Sattath & Tversky (1976) show that it is a testable implication of the independent random utility model and also of the Elimination by Aspects (EBA) model described in Tversky (1972*a*) and Tversky (1972*b*). McCausland & Marley (2024) show that any system of binary probabilities can
5 be extended to a complete random choice structure in such a way that the complete random choice structure satisfies the multiplicative inequality. As with regularity, each $P_A(x)$ is pivotal.

The conditions we consider here are just three representative examples. We direct the reader to Fishburn (1999), which documents these and many conditions on choice probabilities, including ones that have been proposed as reasonable assumptions about choice behaviour and some that are
10 testable implications of different classes of random choice models. See Luce & Suppes (1965) and McCausland & Marley (2013) for illustrations of the logical relationships among a set of conditions that includes the three outlined here and various kinds of stochastic transitivity.

2.3 Individual- and population-level choice

Most experiments involve a sequence of choice trials, where some menu is presented at each trial.
15 We distinguish two different types of repeated choice behaviour. We will qualify choice behaviour as *individual*-level when it consists of the repeated choices of a single individual and *population*-level when a participant is chosen at random from a population at each trial. Hybrid experiments are possible, but we are not concerned with these here. We have seen that our experiment has a population-level design.

A random choice model on some universe can be used to describe individual- or population-
20 level behaviour, but we should not expect the same model to apply to both. For example, we would expect more heterogeneity in population-level choice, and thus fewer choice probabilities close to zero or one. Whether or not a condition like regularity applies to individual-level choice and whether or not it applies to population-level choice may both be of interest, but these are two
25 distinct scientific questions.

That said, the two questions are sometimes related, through the following aggregation property: if the random choice model for all individuals in a population satisfies the condition, the model describing the population also satisfies the condition. Expressed in contrapositive form to make plain the implications for testing, if the condition is violated for a population, then it must be

violated by at least one individual.

Some, but not all conditions have this aggregation property. Fortunately, a simple mathematical test tells us whether or not any given condition does or does not. Define Δ as the set of all random choice models and Λ as the subset of Δ where a given condition of interest holds. Then the condition satisfies the aggregation property if and only if Λ is a convex set. This is because the population random choice model is a mixture—where random sampling from the population provides the mixing distribution—of the various individual random choice models. If all individual random choice models lie in the same convex set, then the mixture, a convex combination, must also. If the condition is not convex, we can find two individual random choice models that satisfy the condition such that the random choice model for the population of two does not.

We can see that the regularity condition satisfies the aggregation property above. The regularity inequalities are all linear in choice probabilities, and so the regularity region is the intersection of the convex set Δ and many half-planes, also convex. Therefore, the region is convex and regularity satisfies the aggregation property. Likewise, the Block-Marschak inequalities are also linear in choice probabilities and so random utility also satisfies the aggregation property. The multiplicative inequality is not linear, and McCausland & Marley (2024) show that it does not satisfy the aggregation property.

Another issue concerning individual- and population-level choice is the assumption, which we maintain throughout, that observed choices are statistically independent and for each menu, identically distributed. There are good reasons to be sceptical of this assumption for individual-level choice. Participants may be learning or their attention may be waning during the course of an experiment. The hypothesis is more plausible for population-level choice, where variation across observations comes from random sampling and not from changes in the behaviour of any individual.

2.4 Context effects

Context effects are designed to elicit anomalous choice behaviour of different kinds. To provide challenging choice environments for stochastic choice conditions, we deliberately design some of the choice domains to try to elicit three well documented “context effects”: the attraction, similarity and compromise effects. Here we review the effects and give some examples. In Appendix B, where we describe the choice domains in detail, we describe how some of the domains (not limited to the examples in this section) are designed to elicit these effects.

The three context effects can be seen as contingent conditions on choice probabilities, conditions

predicted to hold when the objects in question are in certain relationships with each other. In the example in the introduction, the condition predicted to hold is that the probability of choosing Vancouver goes up when Seattle is added to the menu. The prediction is contingent on Seattle being clearly “inferior to” (i.e. obviously further south than) Vancouver. We will describe seminal
5 versions of each of the three effects. Other versions of these context effects have been proposed, and they have different aggregation properties and different implications on whether choice probabilities can satisfy regularity, the Block-Marschak inequalities, or other conditions not considered here, such as Krantz’s (1967) simple scalability. See Davis-Stober et al. (2023) for a comprehensive discussion.

The attraction effect (sometimes known as the asymmetric dominance effect) was introduced by
10 Huber et al. (1982), and pertains to choice environments with three objects and some dominance relation over these objects. In one experimental design from Huber et al. (1982), objects are six-packs of beer with two attributes, price and quality rating. A six-pack dominates another when its price is lower and its quality higher at the same time. The three objects are a “target” x , a “competitor” y and a “decoy” z ; x dominates z , but y does not; neither x nor y dominates
15 the other. In the six-pack example, the target x has a price of \$1.80 and a rating of 50. The competitor y has a higher price (\$2.60) but also a higher rating (70) than the target, so neither dominates the other. The decoy z has a higher price (\$2.20) than the target and a lower rating (40). The target dominates the decoy because it is cheaper and has a higher rating. Importantly, the competitor does not dominate the decoy; this establishes an asymmetry between the target and
20 the competitor. The effect is said to occur when the probability of choosing the target goes up when the decoy is added to the pair of target and competitor; that is, when $P_{\{x,y,z\}}(x) > P_{\{x,y\}}(x)$. This condition constitutes evidence against regularity, since one of the regularity inequalities is $P_{\{x,y,z\}}(x) \leq P_{\{x,y\}}(x)$, and random utility, because regularity is a necessary condition for random utility. The “Beer” domain of our experiment is based on this design, but includes five objects (and
25 prices updated from 1982).

We also include domains where the dominance relation is not based on explicit levels of attributes. We have seen one example from the “Latitude” domain, a configuration with Vancouver as target, Warsaw as competitor and Seattle as decoy. Two other configurations feature London and Paris as competitor in place of Warsaw.

30 Tversky (1972*b*) introduced the similarity effect, also pertaining to choice environments with three objects. Here, there is a similarity relation among the choice objects under consideration. Our “Dots” domain is similar to one of the three experimental designs in Tversky (1972*b*) and includes

boxes with 320, 310, 300, 290 and 280 points, labelled a , b , c , d and e , respectively. The prompt question is “Which one of the following boxes do you think has the greatest number of points?”

We can define a similarity relation that considers two adjacent boxes in this list as similar. The similarity effect occurs if there are objects x , y and z such that x and y are similar, x and z are

not similar and y and z are not similar and choice probabilities satisfy

$$P_{\{x,z\}}(x) > \frac{P_{\{x,y,z\}}(x)}{P_{\{x,y,z\}}(x) + P_{\{x,y,z\}}(z)} \quad \text{and} \quad P_{\{y,z\}}(y) > \frac{P_{\{x,y,z\}}(y)}{P_{\{x,y,z\}}(y) + P_{\{x,y,z\}}(z)}.$$

The inequality on the left holds if adding the object y , similar to the target object x , to the menu $\{x, z\}$ lowers the *relative* probability of choosing x , compared to z . The inequality on the right is similar, except that y is the target object and x is the similar object. The tripleton set $\{a, b, e\}$ is a candidate for the similarity effect, with a and b as the similar elements and e as the dissimilar

element. Likewise, $\{a, d, e\}$ is also a candidate, with d and e as the similar elements.

Simonson (1989) introduces a strong form of the compromise effect and Tversky & Simonson (1993) introduce a weaker version. They pertain to choice environments where objects can be described in terms of a set of real-valued attributes. An object y is *between* objects x and z if two conditions hold: first, for each attribute, y has a level of that attribute intermediate between the levels of x and z ; and second, that each of x and z have the highest level of at least one attribute. The weak effect is said to occur when y is between x and z and the following inequalities, from page 1183 in Tversky & Simonson (1993), hold:

$$P_{\{y,x\}}(y) < \frac{P_{\{x,y,z\}}(y)}{P_{\{x,y,z\}}(y) + P_{\{x,y,z\}}(x)} \quad \text{and} \quad P_{\{y,z\}}(y) < \frac{P_{\{x,y,z\}}(y)}{P_{\{x,y,z\}}(y) + P_{\{x,y,z\}}(z)}.$$

The inequality on the left holds if adding the object z to the menu $\{x, y\}$, making y a “compromise” between x and z , raises the relative probability of choosing y , compared to x . The inequality on the right is similar, except that x is added to $\{y, z\}$, again making y a “compromise” between x and z . Our “Beer” domain includes three six-packs where a betweenness relation, relative to the price and quality rating attributes, applies.

The similarity and compromise effects are compatible with the conditions we consider in this paper, but not with others that may be of interest, such as the constant ratio rule described by Tversky (1972a). See Davis-Stober et al. (2023) for details; different versions of context effects may differ in terms of their compatibility with different conditions.

3 Experimental design

Here we describe the experiment, pointing out design features that serve the objective of testing abstract stochastic choice conditions across a wide variety of choice environments.

3.1 Selection of choice domains

5 An important objective is to cover a wide variety of choice environments. For this reason we chose domains to span several qualitatively different kinds of choice, including consumer choice, aesthetic preference, factual judgment and perceptual discrimination. For example, the “Beer” domain mentioned above mimics domains used in consumer choice experiments where different objects have different levels of various attributes. In the “Juice” domain, participants choose among
10 different kinds of juice. In some domains, participants make judgments, such as the “Sentences” domain, where they make relative judgments about grammaticality. In some domains, like the “Dots” domain described above, choices involve perception. Our experiment features 32 choice domains in total, each domain consisting of a universe of five related choice objects.

Table 2 shows the names of the choice domains, together with the prompt (or an excerpt of the
15 prompt, with excluded text indicated by an ellipsis) used to elicit a response from the participant. Appendix B describes each domain in more detail, giving the full prompt, describing the five choice objects and, where applicable, providing a source for the information used to construct the question. Figure 1 shows screenshots for the “Travel” and “Coffee” domains. The screenshot for the “Travel” domain shows a menu with two of the five choice objects in that domain. The screenshot for the
20 “Coffee” domain shows a menu with four of the five choice objects in that domain.

To better stress-test stochastic conditions, we ensure that domains intended to elicit anomalous behaviour, in the form of context effects, are well represented. Some examples we have seen above are the “Latitude”, “Beer”, and “Dots” domains. Section B describes these and several others in detail.

25 The variety of domains may also have contributed to unusually high participant engagement, as reflected in voluntary comments submitted at the end of the survey, available in the Supplementary Materials.

Travel

Which one of the following travel destinations would you most like to visit?


Istanbul, Turkey


Marrakech, Morocco

>>

Coffee

You need to buy 16oz of ground coffee for a brunch with friends. Which one of the following ground coffees would you choose?

	Price (\$)	Fair Trade	Name: Description
☐	18.71	Yes	Ethiopian Yirgacheffe: vibrant and intensely aromatic, fruity
☐	9.99	No	Colombian Supremo: mellow cup, complex aromas and rich flavours
☐	13.72	Yes	Colombian Organic: medium body, fragrant aroma and mild acidity
☐	13.46	No	Sumatra Mandheling: exotic, earthy, bright with low acidity

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Figure 1: A screenshot for the “Travel” domain, in which participants select pictures, and a screenshot for the “Coffee” domain, in which participants select radio buttons. In the experiment and in electronic versions of this paper, these photographs are in colour.

Table 2: Choice domains of the experiment, with prompt question

Domain name	Question
Male stars	Which movie star would you choose to have lunch with?
Female stars	Which movie star would you choose to have lunch with?
Films	Judging from the following descriptions of films, which one of the films would you choose to see?
Star pairs	Knowing only who is starring, which one of these new films would you choose to see?
Pizzas	Which one of the following pizzas would you choose?
Juices	Which one of the following fresh juices would you choose?
Colours	Which one of the following colours do you like best?
Colour combos	Which one of these colour combinations do you like best?
Events	Which one of the following events do you think is most likely to happen in the next twenty years?
Radio formats	Suppose you were on a two hour road trip and you have a choice among radio stations with the following formats. Which one would you choose?
Music	Which one of the following musical artists do you like the best?
Aboriginal	Which one of the following examples of Australian aboriginal art do you like the best?
Impressionist	Which one of the following examples of Impressionist art do you like the best?
Sentences	Which one of the following sentences do you find the most grammatically acceptable?
Travel	Which one of the following travel destinations would you most like to visit?
Marijuana	Which one of the following marijuana policies would you choose?
Latitude	Which one of the following cities do you think is furthest north?
Dots	Which one of the following boxes do you think has the greatest number of points?
Triangles	Which one of the following triangles do you think has the greatest area?
Population	Which one of the following countries do you think has the largest population?
Surface area	Which one of the following countries do you think has the greatest surface area, including inland bodies of water?
Beer	... Given that you had to choose one brand to buy on this information alone, which one would you choose?
Cars	Which one of the following cars would you choose to drive, all other features being equal? ...
Restaurants	Which one of the following restaurants would you choose for your next restaurant meal...?
Layovers	Which one of the following flight itineraries would you choose? ...
Future	Which one of the following would you choose?
Phone plans	Of the following cell phone plans, which one would you choose? ...
Hotel rooms	... Which one of the following hotels would you choose, ...?
Itineraries	Which one of the following flight itineraries would you choose? ...
Televisions	Which one of the following televisions would you choose to buy ...?
Coffee	Which one of the following ground coffees would you choose?
Charity	Suppose someone was donating a total of 100 dollars to a combination of charities, on your behalf. Which one of the following divisions of the 100 dollars would you choose?

3.2 A population-level full-menu design

Many of the stochastic choice conditions one might want to test have the property that for each menu A , the probability vector $P_A(\cdot)$ is pivotal, in the sense defined in Section 2. This is true for the

three conditions we have considered in this paper. Because observed choices from any menu have the potential to provide evidence against these conditions, it is important to observe choices from every menu of a domain. We have settled on a purely population-level design, to ensure that for each domain, choices are plausibly independent and, menu by menu, identically distributed. For these reasons, each participant sees exactly one menu from each domain, exactly once; and each menu of each domain is seen by several participants.

Here are the details of the design. There are $J = 32$ choice domains, each with a 5-object universe. For each domain, there are $I = 2^5 - 5 - 1 = 26$ menus (binary and larger subsets of the five-element universe), for a total of $J I = 832$ menus. Since each participant faces exactly one menu from each domain, once, the number of choice trials for each participant is the same as the number of domains, $J = 32$.

Our experimental design assigns participants to menus as follows. For each domain, each of the $I = 26$ menus of that domain is presented $N = 40$ times, once to each of N distinct participants. So for the “Colours” domain, for example, 40 different participants see the menu `{Green,Blue}` and another 40 see the menu `{Blue,Red,Purple}`, and so on for all the menus of that domain. With $I = 26$ menus per domain, we have $N I = 40 \cdot 26 = 1040$ distinct participants, each choosing a colour from some menu of colours. For any other domain, we observe the same 1040 participants choosing one object from some menu of that other domain and again, each of the 26 menus of the other domain is seen by 40 different participants.

Thus the total number of trials is $N I J = 40 \cdot 26 \cdot 32 = 33,280$, which we can factor in different ways. Each of the $N I = 1040$ participants sees one menu from each of the $J = 32$ domains for a total of $N I \cdot J$ trials. Each of the $J = 32$ domains has $I = 26$ menus and each of these $J I = 32 \cdot 26 = 832$ menus is seen by $N = 40$ participants, for a total of $J I \cdot N$ trials.

This was the intended design, but in the event, two extra participants completed the experiment, and their choices were included in the analysis. So for each domain, exactly two menus were presented 41 times and the remaining 24 menus were presented 40 times, as planned. Thus the actual total number of trials is 33,344.

3.3 Other design features

Participants are constrained to choose exactly one object from each menu. For some domains, participants select radio buttons and in others, they select pictures, which are then identified by a border. Figure 1 shows one screenshot for each kind of domain. In each case, participants can

modify their choice before they confirm it by clicking on a button labelled “>”; once they click this button, they cannot go back and change their response. A progress bar indicates progress through the experiment.

Assignments of participants to menus and the positions of menu items on the screen are random.

5 For each domain, the participants are randomly partitioned into $I = 26$ groups of size $N = 40$, with participants in each group seeing the same menu from that domain. Random partitions are uniformly distributed and mutually independent across domains, so it is unlikely that any two participants will be in the same group and thus see the same menu more than a couple of times. Elements of a menu are presented in random position (left to right or top to bottom, as the case
10 may be) on the screen, independently across the N participants facing that same menu. The J trials assigned to each participant are presented in random order, independently across participants, so it is exceedingly unlikely that two participants see the domains in the same order. Random partitions, positions and trial order are mutually independent.

3.4 The sample of participants

15 The experiment consisted of an internet survey conducted between August 10, 2017 and August 31, 2017. A total of 1042 participants completed the survey. Participants were recruited by Survey Sampling International (SSI) from their Canadian panel, designed to be representative of the Canadian adult population. SSI screens potential members for reliability and requires that all participants be at least 18 years of age. It also verifies the identity of their participants for each
20 survey. Compensation of participants includes cash and other rewards. The Supplementary Materials include copies of the consent form, a Certificate of (ethics) Approval, the SSI Global Panel Book 2017, and a document describing in more detail the recruitment and compensation procedures followed by SSI. Programming and hosting of the survey were provided by the Institute for Choice at the University of South Australia.

25 We chose not to impose any additional restrictions on our sample of participants. We have full information on age, province or territory, and sex for all 1042 participants in our sample. There are 515 men (49.4%) and 527 women (50.6%). Age ranges from 18 to 88; the percentage of participants in the age bracket 18-24 is 11.4%; 25-34, 17.8%; 35-54, 38.3%, 55 and older, 32.5%. The number of participants in Alberta was 124; British Columbia, 136; Manitoba, 36; New Brunswick,
30 23; Newfoundland and Labrador, 16; Northwest Territories, 1; Nova Scotia, 29; Ontario, 391; Prince Edward Island, 4; Quebec, 251; Saskatchewan, 30; and Yukon, 1. Participants could take

as long as they liked to complete the survey. The quartiles of total duration were 7.1, 10.0 and 13.9 minutes. Three participants took longer than 24 hours, and the shortest duration was 1.7 minutes. Participants were given the opportunity to leave written feedback on the experiment upon completion. A complete list of participants’ feedback is in the Supplementary Materials.

4 Bayesian models and inference

In Section 2, we defined a random choice model P on a universe U of objects. The assumption that choices are independent, and for each menu identically distributed, completes the description of the data generating process for experimental data like ours. We will denote by N the experimental data as a random vector, before observation. We will denote by n a generic possible value of the data vector and by n_{obs} the realized data vector we observe. The data generating process gives the conditional distribution $N|P$.

We use Bayesian inference to learn about plausible values of the unknown P after observing N . Because the goal is to test stochastic choice conditions, we conduct inference directly on the space of all random choice models and not a reduced-dimension parametric class. That said, because there are a finite number of menus and options, P is a finite dimensional parameter vector. For a universe of size five, there are 75 choice probabilities subject to 26 adding-up constraints, leaving 49 free parameters. The parameter space, which we denote by Δ , consists of all values of the vector P that satisfy the axioms of probability.

Bayesian inference requires a prior distribution for P . Once we have this, we have a joint distribution of P and N and we can derive, in principle, the posterior distribution of P (i.e. the conditional distribution of P given N) and the marginal distribution of N . We will call the unknown random choice model P , completed by a prior distribution over P , a *Bayesian random choice model*. The use of the word “model” here is consistent with its use in the term “random choice model”. Just as a particular known random choice model describes an unconditional distribution for N , a Bayesian random choice model implies an unconditional distribution for N .

Our main inferential objective is testing stochastic choice conditions using data, which in a Bayesian context is an exercise in model comparison. To test a given condition, we will compare two Bayesian random choice models, an *encompassing* model, where no further restrictions are placed on P , with a constrained model, where P is restricted to the set of random choice models that satisfy the condition. To do this we will use two different kinds of prior distribution. The

encompassing model is a Bayesian random choice model where the prior distribution of P has full support on Δ , to ensure that it does not rule out any random choice models *a priori*. The constrained model is a Bayesian random choice model where the prior distribution is the truncation of the prior distribution of the encompassing model to the region where the condition holds.

5 The standard summary statistic for a comparison of two Bayesian models is the Bayes' factor. If two Bayesian random choice models, M and M' , are both considered equally likely before seeing the data, the Bayes factor in favour of M' over M gives the posterior odds ratio $\Pr[M'|N = n_{\text{obs}}]/\Pr[M|N = n_{\text{obs}}]$. So for example, if the two models are equally likely to begin with and the Bayes factor is 2, the posterior assigned to model M' is twice as high as the posterior probability
10 assigned to M .

The Bayes factor is defined as a ratio of marginal likelihoods:

$$\text{BF} \equiv \frac{f_{M'}(n_{\text{obs}})}{f_M(n_{\text{obs}})},$$

where $f_M(n)$ is the marginal probability mass function of the data vector under model M , and $f_{M'}(n)$ is the same under M' . Evaluating $f_M(\cdot)$ at the observed data n_{obs} gives a measure of the prior predictive performance of M for the observed data. Because it averages over all parameter
15 values rather than evaluating a model's predictive performance at a fitted estimate, it measures predictive performance on unseen data.

For the special case when M is an encompassing model and M' is a constrained model obtained by truncating the prior distribution of M to the region Λ of Δ where a condition of interest holds, the Bayes factor reduces to the following intuitively satisfying form:

$$\text{BF} = \frac{\Pr_M[P \in \Lambda | N = n_{\text{obs}}]}{\Pr_M[P \in \Lambda]}. \quad (2)$$

20 The numerator, the posterior probability that the condition holds, under the encompassing model, is a measure of the compatibility of the condition with the observed data. The denominator, the prior probability that the condition holds, under the encompassing model, is an inverse measure of model parsimony. The lower the denominator is, the more restrictive the condition. See Klugkist & Hoijtink. (2007) for details. For further explanation of Bayes factors, see Berger (1985) or Bernardo
25 & Smith (1994).

The ratio in (2) also suggests a practical way of computing approximations of Bayes factors. Since the probability of an event is just the expectation of an indicator function for the event, we

Target	Competitor	Decoy	$(n_T^{(2)}, n_C^{(2)})$	$(n_T^{(3)}, n_C^{(3)}, n_D^{(3)})$	$\Pr[P \in \Lambda N = n_{\text{obs}}]$	BF	ln BF
Vancouver	Warsaw	Seattle	(9, 31)	(9, 30, 2)	0.453	1.359	0.306
Vancouver	London	Seattle	(11, 29)	(15, 22, 3)	0.805	2.415	0.882
Vancouver	Paris	Seattle	(17, 23)	(26, 14, 0)	0.969	2.907	1.067
London	Warsaw	Paris	(18, 22)	(14, 24, 2)	0.162	0.486	-0.722
London	Vancouver	Paris	(29, 11)	(24, 14, 2)	0.096	0.289	-1.240
London	Seattle	Paris	(31, 9)	(25, 7, 8)	0.056	0.169	-1.780

Table 3: Data and test results for six candidate attraction effects in the Latitude domain. Each row gives a configuration of target, competitor and decoy options where we might expect an attraction effect. The fourth column gives the numbers of participants choosing the target and competitor when the menu is this pair of options. The fifth column gives the numbers of participants choosing the target, competitor and decoy when the menu is the set of these three options. The sixth column gives the posterior probability of the hypothesis that the attraction effect holds. The last two columns give the Bayes Factor and log Bayes Factor in favour of the hypothesis that the attraction effect holds.

can estimate numerator and denominator using Monte Carlo methods. In the next section we turn to some empirical analyses, reporting Bayes factors estimated numerically in this way in order to compare various Bayesian random choice models.

5 Empirical Illustrations

We begin with a simple context effect analysis involving the comparison of just two choice probabilities. This illustrates the Bayesian testing framework in its simplest form. We then perform global tests of the three stochastic choice conditions outlined in Section 2. We end by showing a graphical representation of all binary and ternary choice proportions for a domain, a representation that can reveal potential violations of regularity and the multiplicative inequality.

5.1 Tests of context effects using Bayes factors

Here we provide a worked example showing how to test whether or not a context effect holds in a given situation. We consider six possible configurations of target, competitor and decoy in the “Latitude” domain, based on the dominance relation that puts Vancouver over Seattle, as proposed in Section 1, and London over Paris. Each row of Table 3 describes one configuration, reports the relevant choice counts and reports test results for the hypothesis that the effect holds. The first three columns identify the target (T), competitor (C) and decoy (D) options that define the configuration. The fourth column gives $n_{\{T,C\}}(T)$ and $n_{\{T,C\}}(C)$, the number of participants

choosing the target and competitor, respectively, from the menu of these two options. The fifth column gives $n_{\{T,C,D\}}(T)$, $n_{\{T,C,D\}}(C)$ and $n_{\{T,C,D\}}(D)$, the number of participants choosing the target, competitor and decoy, respectively, from the menu of these three options. We see that the number of people choosing Vancouver goes up in the presence of Seattle, the decoy, when the competitor is either London or Paris. This is suggestive of an attraction effect.

Observations from menus consisting of the proposed target and decoy options provide some independent evidence on the plausibility of the two dominance relations, and these suggest that Seattle is a more convincing decoy for Vancouver than Paris is for London. Of the participants who saw the menu of Vancouver and Seattle, 36 chose Vancouver and 4 chose Seattle. Of those who saw the menu of London and Paris, only 32 chose London and 8 chose Paris.

For each configuration, the hypothesis is that the probability of choosing the target increases when the decoy is added to the menu. The analysis therefore reduces to inference about two unknown probabilities and whether one exceeds the other. We use an encompassing model with a prior distribution where all probability vectors are independent and uniformly distributed. In particular, the probability vectors of interest, $(P_{\{T,C\}}(T), P_{\{T,C\}}(C))$ and $(P_{\{T,C,D\}}(T), P_{\{T,C,D\}}(C), P_{\{T,C,D\}}(D))$ are independent and each is uniformly distributed on the region where its elements are non-negative and add to one. The hypothesis we wish to test is that the attraction effect holds, or equivalently, that $P_{\{T,C\}}(T) < P_{\{T,C,D\}}(T)$. Taking Λ to be the region of Δ where $P_{\{T,C\}}(T) < P_{\{T,C,D\}}(T)$, the constrained model has a prior distribution for P that is the truncation of the prior distribution in the encompassing model to the region Λ .

We use standard results for the Dirichlet-multinomial and beta-binomial models (see for example Gelman et al. (1995)) to analytically derive the posterior distribution of $P_{\{T,C\}}(\cdot)$ and $P_{\{T,C,D\}}(\cdot)$. In the following derivation, we denote by $\text{Be}(\alpha, \beta)$ the beta distribution with shape parameters α and β . The prior independence of $P_{\{T,C\}}(\cdot)$ and $P_{\{T,C,D\}}(\cdot)$ and the assumed independence of participants' choices imply the posterior independence of these probability vectors. The uniform distributions of $P_{\{T,C\}}(\cdot)$ and $P_{\{T,C,D\}}(\cdot)$ imply that the prior distribution of $P_{\{T,C\}}(T)$ is $\text{Be}(1, 1)$ and that that of $P_{\{T,C,D\}}(T)$ is $\text{Be}(1, 2)$. Then the posterior distribution of $P_{\{T,C\}}(T)$ is $\text{Be}(1 + n_{\{T,C\}}(T), 1 + n_{\{T,C\}}(C))$ and that of $P_{\{T,C,D\}}(T)$ is $\text{Be}(1 + n_{\{T,C,D\}}(T), 2 + n_{\{T,C,D\}}(C) + n_{\{T,C,D\}}(D))$.

The test for the attraction effect is based on the Bayes factor in favour of the constrained model over the encompassing model, computed as $\Pr[P \in \Lambda | N = n_{\text{obs}}] / \Pr[P \in \Lambda]$, the ratio of posterior to prior probability of the effect under the encompassing model. We computed $\Pr[P \in \Lambda] = \frac{1}{3}$ analytically. We estimated $\Pr[P \in \Lambda | N = n_{\text{obs}}]$ numerically with Monte Carlo simulation methods,

using Rao-Blackwellization to reduce variance, and computed numerical standard errors.

Column seven of Table 3 give numerical estimates of $\Pr[P \in \Lambda | N = n_{\text{obs}}]$, the posterior probability of the attraction effect under the encompassing model. All numerical standard errors are less than 0.0001. The final two columns give the Bayes Factor (BF), equal to the ratio
 5 $\Pr[P \in \Lambda | N n_{\text{obs}}] / \Pr[P \in \Lambda]$, and its natural logarithm.

The evidence favours the attraction effect for the three cases where the target is Vancouver and the decoy is Seattle. In contrast, it favours the encompassing model for the three cases where the target is London and the decoy is Paris.

We can use the guidelines recommended by Kass & Raftery (1995) to classify the strength of
 10 the evidence provided by the Bayes Factors. They classify log Bayes factors between 0 and 1 as “not worth more than a bare mention”, those between 1 and 3 as providing “positive evidence”, those between 3 and 5 as providing “strong evidence” and those above 5 as providing “very strong evidence”. We will adopt these categories throughout the paper, but refer to the first two as giving “weak” and “moderate” evidence.

15 By this reckoning, there is moderate evidence in favour of the attraction effect when the target is Vancouver, the competitor is Paris and the decoy is Seattle and moderate evidence against the attraction effect when the target is London, the decoy is Paris and the competitor is either Vancouver or Seattle.

We can also consider joint hypotheses, such as the hypothesis that the attraction effect holds
 20 in all three cases where the target is Vancouver and the decoy is Seattle. The log Bayes factor in favour of this joint hypothesis is the sum of the individual log Bayes factors, or 2.253. This is stronger evidence, but still not “strong” according to the standards of Kass & Raftery (1995). The log Bayes factor in favour of the joint hypothesis that the attraction effect holds in all three cases where the target is London and the decoy is Paris is -3.744, which implies “strong” evidence
 25 against the joint hypothesis.

5.2 Tests of three global abstract conditions

We have seen that regularity, the multiplicative inequality and the Block-Marschak inequalities are also joint hypotheses. However, the number of inequalities involved is much greater. Regularity, for example, consists of 70 distinct inequalities when the universe consists of five objects.

30 It is natural to choose a prior distribution where the $P_A(\cdot)$ are mutually independent across menus. We did this above when we tested for multiple attraction effects. Cavagnaro & Davis-Stober

(2014), Davis-Stober et al. (2015), Myung et al. (2005) and Zwillling et al. (2011) do the same for testing simpler and less restrictive conditions than the three we investigate here.

Although the simplicity of such a prior is in its favour, some kinds of prior correlation between choice probabilities on different menus are intuitively compelling. Suppose you observe choices only
5 from the menu $\{a, b, c\}$. You don't know what the objects a , b and c are but see that many more people choose a than b . Now you appraise choice probabilities on the smaller set $\{a, b\}$. We suspect most people would put more weight on higher values of the probability $P_{\{a,b\}}(a)$ than they would before observing choices from $\{a, b, c\}$, despite not knowing what the objects are and having no direct evidence in the form of choices from $\{a, b\}$.

10 Furthermore, computational problems arise when we test our three very restrictive conditions using a prior distribution where the $P_A(\cdot)$ are independent. For a universe of five objects, the prior probabilities of these conditions are so tiny that the amount of simulation required to measure them using straightforward simulation would be prohibitively high.

McCausland & Marley (2013) address both these issues by introducing a prior distribution for P
15 where the $P_A(\cdot)$ are dependent. For testing abstract choice conditions, the prior has some attractive properties, such as invariance to relabeling objects. The prior has two parameters, $\lambda \in [0, 1]$ and $\alpha > 0$.

The λ parameter controls the degree of dependence of choice probabilities across menus. When $\lambda = 0$, the $P_A(\cdot)$ are mutually independent. Thus, if you adopt a prior where $\lambda = 0$, you learn
20 nothing about $P_{\{a,b\}}(a)$, even if you observe a thousand people out of a thousand choosing a from $\{a, b, c\}$. As λ increases, the prior probability of regularity, the multiplicative inequality and the Block-Marschak inequalities all increase. The marginal distribution of each $P_A(\cdot)$ does not depend on λ , but correlations among them do. For example, the correlation between $P_{\{a,b,c\}}(a)$ and $P_{\{a,b\}}(a)$ increases in λ , even though their two marginal distributions do not depend on λ . The
25 prior is constructed so that at the maximal value $\lambda = 1$, the probabilities of regularity and the Block-Marschak conditions are both one.

The α parameter governs the degree to which choice probabilities concentrate near zero or one, with lower values of α corresponding to more probability mass near zero or one. Thus, low values of α are associated with more extreme choice probabilities. High values of α are associated with
30 less extreme choice probabilities. For α greater than the number of elements in a menu A , there is a prior mode at the uniform discrete distribution on A , where all objects are equally likely to be chosen. As α increases, more prior probability mass concentrates near this mode.

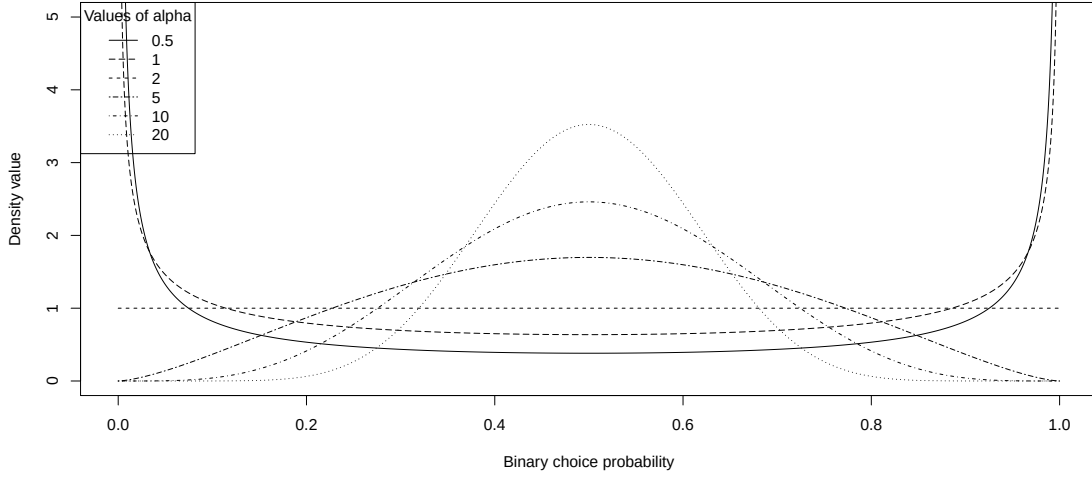


Figure 2: Density of binary choice probabilities for $\alpha = 0.5, 1, 2, 5, 10, 20$. All densities are symmetric Beta with both shape parameters set to $\alpha/2$

Figure 2 illustrates how the prior distribution of any binary choice probability varies with α . For $\alpha < 2$, the density is unbounded, approaching infinity as the probability gets close to zero or one. For $\alpha = 2$, the probability is uniformly distributed on the interval $[0, 1]$. For $\alpha > 2$, there is a mode at $\frac{1}{2}$, and as α increases, the variance decreases.

While we use the same family of priors here that we did for previous analyses in McCausland & Marley (2014) and McCausland et al. (2020) some details are different. The main reason is that the previous papers study individual-level choice and the current paper studies population-level choice. We expect heterogeneity across individuals. Therefore we anticipate more choice variability and less extreme choice probabilities at the population level. For this reason, we favour higher values of α and therefore less prior probability mass near zero and one, compared with the previous papers.

It is possible to specify a prior distribution for P by choosing values for α and λ , but this is quite inflexible. Instead, we specify a prior distribution for them, completing a hierarchical prior for P . Because results are somewhat sensitive to the choice of prior for λ (although less so to the choice of α) we do a prior sensitivity analysis. Specifically, we perform tests of each condition using three different encompassing models, M_1 , M_2 and M_3 , which differ only in terms of the prior distribution of λ . M_1 favours lower values of λ than M_2 and M_3 favours higher values of λ . Thus, the three models correspond roughly to weak, moderate and strong prior dependence across menus. To allow exact reproducibility, we provide details of the three prior specifications in Appendix A.

For each choice domain, we computed $\log^1$ marginal likelihoods for the three encompassing models, M_1 , M_2 and M_3 , and for the constrained models defined by all possible pairs of encompassing model and condition (regularity, the multiplicative inequalities or the Block-Marschak inequalities).

Recall that the log marginal likelihood summarizes the prediction record of a model, and the log Bayes factor for comparing any two models is the difference between their log marginal likelihoods.

We report log Bayes factors for all twelve models relative to the encompassing model M_3 , to establish a reference level of zero on a scale of predictive performance; by definition, the log Bayes factor for M_3 relative to itself is zero. We choose M_3 as the reference model because it was the most favoured (measured by log marginal likelihood) encompassing model for 31 out of the 32 domains.

All other Bayes factors can be computed as differences of these: a Bayes factor is the difference between the marginal likelihoods of two models, also equal to the difference between their respective Bayes factors relative to the *same* reference model, here M_3 .

Figure 3 gives, for each choice domain, log Bayes factors for various combinations of condition and encompassing model, relative to encompassing model M_3 . There is a panel for each domain, and each panel shows three columns of points, corresponding to the three encompassing models M_1 , M_2 and M_3 . The vertical scale gives Bayes factors relative to M_3 ; the choice of M_3 as the reference model just establishes a zero on the scale. Bayes factors between any two models out of twelve can be read off as vertical distances between two points. In particular, the vertical distance between a symbol and the circle in the same column i , $i = 1, 2, 3$, is the log Bayes factor in favour of the corresponding condition for the encompassing model M_i . So for example, the vertical distance between the triangle and the circle in column i gives the Bayes factor in favour of regularity, for the encompassing model M_i .

Each column i , $i = 1, 2, 3$ shows, on this vertical scale, the Bayes factor for the (unconstrained) encompassing model M_i , and the Bayes factor of the three constrained models based on M_i , all relative to M_3 . The circle gives the Bayes factor for the encompassing model M_i itself, relative to M_3 . The triangle, square and cross give log Bayes factors of the constrained model consisting of the restriction of M_i to the regions Λ_{Reg} , Λ_{RU} and Λ_{MI} , respectively, all relative to the reference model M_3 .

Recall that variation in results among the three encompassing models is used for prior sensitivity analysis. Comparisons of Bayes factors across columns illustrate how results depend on changes to the prior specification.

¹All logarithms are natural (base e) logarithms.

There are no crosses for the choice domains “Latitude”, “Itineraries I” and “Payoffs”, regardless of the encompassing model. This is because the posterior probability of the multiplicative inequality is too close to zero to measure, implying very strong evidence against that condition for those domains. Evidence in support of the multiplicative inequality varies a lot across choice domains, from moderate evidence in favour to very strong evidence against. In each case, evidence is fairly robust to the choice of encompassing model.

The evidence for regularity and random utility (the Block-Marschak inequalities are necessary and sufficient for random utility) tends to be weaker, but more consistent across choice domains. In most cases, the log Bayes factors either favour random utility and regularity over the encompassing models, or are very close to zero, representing little evidence either way. Across domains and encompassing models, the evidence for random utility is generally similar to that for regularity, which is a necessary condition for random utility. When the two differ, the evidence for regularity is usually slightly stronger, indicating some evidence against the higher order features of random utility, but this evidence is quite weak. For individual domains, the evidence in favour of random utility and regularity are at best slight, but collectively, the evidence is much stronger: the log Bayes factor in favour of regularity holding jointly across domains is the sum over domains of the log Bayes factors in favour of regularity, and likewise for random utility.

Figure 3 also allows us to compare two different ways of generating structure in choice probabilities: imposing a condition such as regularity or random utility, and increasing prior dependence across menus through λ . We see that the variation in predictive performance of random utility across encompassing models (height of the square, across columns, for a particular domain) is much less than the variation in predictive performance across the encompassing models themselves (height of the circle, across columns, for the same domain). Increasing λ induces coherence across menus. So does imposing random utility, in its own way. The results suggest that some of the predictive gains from imposing random utility can be replicated by increasing λ . The same observation applies to regularity.

5.3 Data visualization for observed choice proportions

The analyses above summarize evidence through Bayes factors. We conclude with a graphical representation of the observed choice proportions on all binary and ternary choice sets of the “Latitude” domain. While purely descriptive and not inferential, the representation can reveal patterns that suggest violations of regularity and the multiplicative inequality. Corresponding

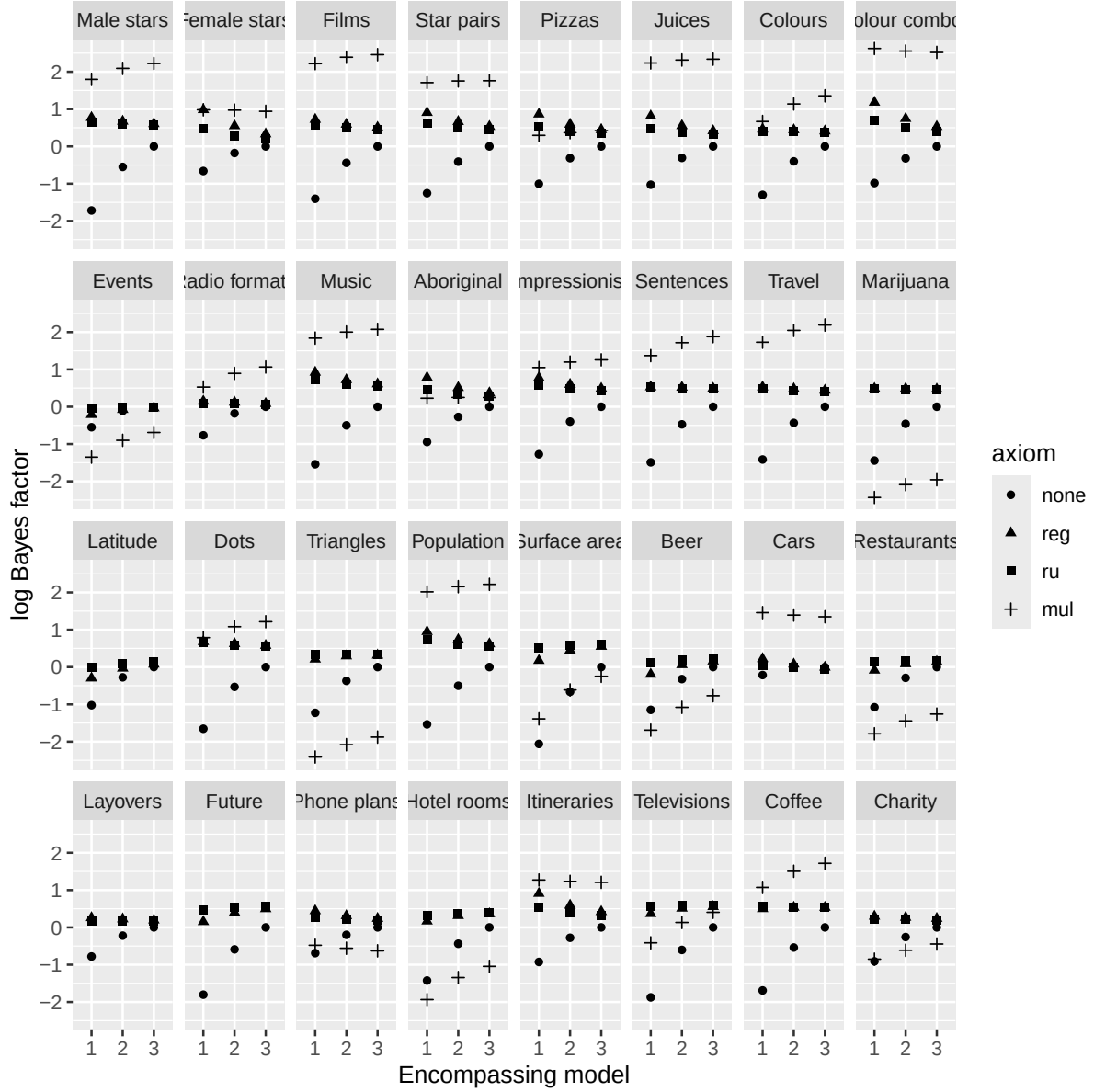


Figure 3: Log Bayes factors for 32 domains, by encompassing model and condition. All log Bayes factors are relative to encompassing model M_3 . The “encompassing model” indices 1, 2 and 3 correspond to models M_1 , M_2 and M_3 . The vertical distance between a symbol and the circle in the same column is the log Bayes factor in favour of the corresponding condition, for a particular domain and encompassing model.

representations for all 32 domains are available in the Supplementary Materials.

We apply a graphical representation introduced by McCausland & Marley (2013) for plotting a point in Δ for a universe of size three, as a set of four points in a Barycentric coordinate system. The point in Δ can be a random choice model specifying choice probabilities for the three binary
5 and one ternary *menus* of the universe, or it can be the corresponding choice proportions observed in a full-menu experiment. The three-object universe can also be a sub-universe of a larger universe of objects.

Recall that for the “Latitude” domain, the universe of objects is $\{a, b, c, d, e\}$ where a is the Warsaw, b is London, c is Vancouver, d is Paris, and e is Seattle. Also recall that Table 1 tabulates
10 observed choice counts and choice proportions for all menus of this domain.

Figure 4 displays ten panels, one for each tripleton subset of $\{a, b, c, d, e\}$. The panel for a given tripleton shows an equilateral triangle with each vertex labelled with one of the elements of that subset. Consider the bottom rightmost panel, for the tripleton $T \equiv \{c, d, e\}$, which we will use as an extended example. This triple corresponds to the configuration of Vancouver (c), Paris (d) and
15 Seattle (e) where we saw the most pronounced evidence for an attraction effect. The vertices of the triangle are labelled c (bottom right), d (top) and e (bottom left). Four menus are subsets of T , $\{c, d\}$, $\{c, e\}$, $\{d, e\}$ and $\{c, d, e\}$; the choice proportions observed for these menus appear in rows 8, 15, 19 and 23 of Table 1, respectively.

We use the graphical representation introduced by McCausland & Marley (2013) to plot the
20 observed choice proportions for these four menus, using an equilateral triangle as a Barycentric coordinate system. It uses a single triangle to display all binary and ternary choice probabilities (or observed proportions) for a universe of three objects. For example, an arbitrary probability vector $(P_T(c), P_T(d), P_T(e))$ on T is represented as the point with Barycentric coordinates $(P_T(c), P_T(d), P_T(e))$ in the triangle. We have represented the vector $(\hat{P}_T(c), \hat{P}_T(d), \hat{P}_T(e)) =$
25 $(\frac{26}{40}, \frac{14}{40}, \frac{0}{40}) = (0.65, 0.35, 0.00)$ of choice proportions observed in the experiment as a solid point. There are two ways of locating this point on the triangle. First, we can take vertices c , d and e as points on the plane of the page or screen. Then the point $(\hat{P}_T(c), \hat{P}_T(d), \hat{P}_T(e))$ is the convex combination of the vertices c , d and e —which have Barycentric coordinates $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$, respectively—with weights 0.65, 0.35 and 0.00. Thus, the solid point is at $0.65c + 0.35d$.
30 The second way relies on the fact that the triangle is equilateral. Taking the triangle to have unit height, the point representing $(\hat{P}_T(c), \hat{P}_T(d), \hat{P}_T(e))$ is at a distance of $\hat{P}_T(c) = 0.65$, to the left side of the triangle, at a distance of $\hat{P}_T(d) = 0.35$ to the base and at a distance of $\hat{P}_T(e) = 0.00$ to the

right side. The upward sloping grid lines, the horizontal grid lines, and the downward sloping grid lines measure out the choice proportions of c , d and e , respectively.

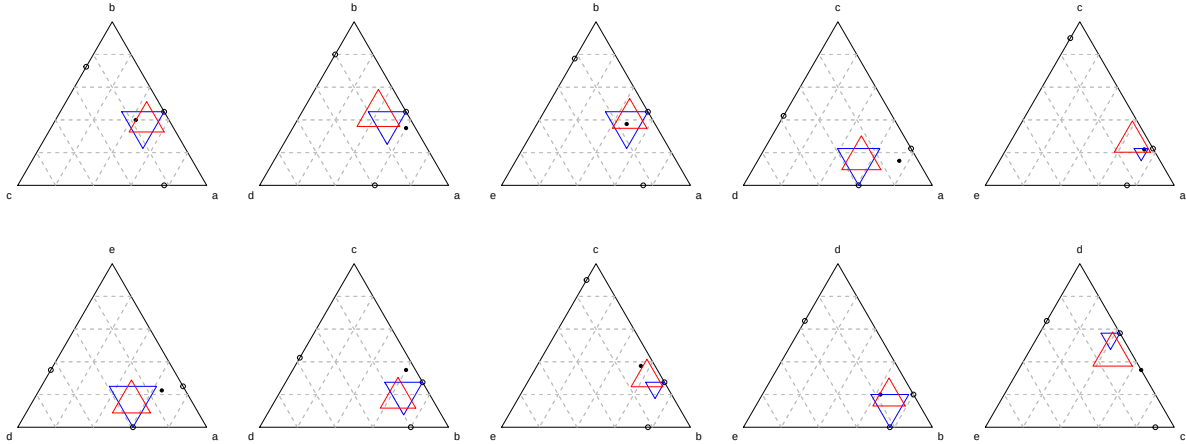
The hollow dots on the right, left and bottom sides of the triangle represent the choice proportions $\hat{P}_{\{c,d\}}(c) = \frac{17}{40} = 0.425$, $\hat{P}_{\{d,e\}}(d) = \frac{26}{40} = 0.65$ and $\hat{P}_{\{(c,e)\}}(c) = \frac{36}{40} = 0.9$, respectively. For example, the hollow dot on the right side of the triangle is the convex combination of the vertices labelled c and d , with weights 0.425 and 0.575 respectively. Equivalently, the distance from this hollow dot to vertex c and the distance to vertex d are in the proportions 0.425 and 0.575.

The two triangles in the interior allow readers to detect, at a glance, if binary and ternary choice proportions for the tripleton set T violate regularity or the multiplicative inequality. Suppose that we know the binary choice proportions $\hat{P}_{\{c,d\}}(c) = \frac{17}{40} = 0.425$, $\hat{P}_{\{d,e\}}(d) = \frac{26}{40} = 0.65$ and $\hat{P}_{\{(c,e)\}}(c) = \frac{36}{40} = 0.90$, but we have not yet observed the ternary choice proportions $(\hat{P}_T(c), \hat{P}_T(d), \hat{P}_T(e))$.

The downward pointing solid triangle with the blue boundary is a cross-section of the region where regularity holds. For the three given binary choice proportions (indicated by the three hollow dots) the solid triangle with the blue boundary is the set of possible ternary choice proportions consistent with regularity and the three observed binary choice proportions. The observed ternary proportions (the solid dot) may or may not lie in the blue triangle. The blue triangle is defined by the regularity conditions on T for proportions, which simplify to $\hat{P}_T(c) \leq \min(\hat{P}_{\{c,d\}}(c), \hat{P}_{\{c,e\}}(c))$, $\hat{P}_T(d) \leq \min(\hat{P}_{\{c,d\}}(d), \hat{P}_{\{d,e\}}(d))$ and $\hat{P}_T(e) \leq \min(\hat{P}_{\{c,e\}}(e), \hat{P}_{\{d,e\}}(e))$. The observed choice proportion vector from menu T , $(\hat{P}_T(c), \hat{P}_T(d), \hat{P}_T(e))$ lies outside the triangle with the blue boundary because $\hat{P}_T(c) = 0.65$ is greater than $\hat{P}_{\{c,d\}}(c) = 0.425$; we conclude that the binary and ternary choice proportions on T do not satisfy regularity. It is important to remember that the observed choice proportions are subject to sampling variation and so the fact that choice proportions violate regularity does not constitute a rejection of the hypothesis that population-level probabilities violate regularity. We are describing here a data visualization method and not the results of statistical tests.

Similarly, the upward pointing solid triangle with the red boundary is a cross-section of the region where the multiplicative inequality holds. For the given binary choice proportions, this solid triangle is the set of ternary choice proportions consistent with the multiplicative inequality and the observed binary choice proportions. The multiplicative inequality on T reduces to $\hat{P}_T(c) \geq \hat{P}_{\{c,d\}}(c)\hat{P}_{\{c,e\}}(c)$, $\hat{P}_T(d) \geq \hat{P}_{\{c,d\}}(d)\hat{P}_{\{d,e\}}(d)$ and $\hat{P}_T(e) \geq \hat{P}_{\{c,e\}}(e)\hat{P}_{\{d,e\}}(e)$. The vector $(\hat{P}_T(c), \hat{P}_T(d), \hat{P}_T(e))$ of ternary choice proportions lies outside the red triangle and in particular,

Figure 4: Observed binary and ternary choice proportions for the “Latitude” domain. In each panel, hollow points represent binary choice proportions and the solid point represents ternary choice proportions. The downward- and upward-pointing triangles indicate the regions consistent with regularity the multiplicative inequality, respectively. Analogous figures for all domains are in the Supplementary Materials.’



$$\hat{P}_T(d) = 0.35 \text{ is less than } \hat{P}_{\{c,d\}}(d)\hat{P}_{\{d,e\}}(d) = 0.575 \cdot 0.65 = 0.37375.$$

6 Conclusions

We have collected new experimental data with observations of population-level choice. Several features of the experimental design make the data suitable for testing abstract conditions on probabilistic choice at the population level. First, it includes a wide variety of choice domains. Some choice domains involve subjective aesthetic judgements, on such matters as food and drink, art, music, politics and the grammaticality of sentences. Other choice domains resemble choice domains that naturally occur in markets, such as those for mobile phone plans, hotel rooms, beer, cars and flights. Still others involve options having an objective rank order. Some choice domains have options that are distinguished only by the levels of numerical attributes, others do not. Some domains are designed to try to elicit context effects, in order to stress-test conditions that are inconsistent with the effects. This variety helps us and others to discover properties of random choice structures that hold widely.

Second, for each choice domain, we observed repeated choices from all menus of a universe of five objects, allowing us to test any condition on probabilistic choice by exposing every implication of it to empirical scrutiny. This is particularly important for testing conditions like regularity, the Block-Marschak inequalities and the multiplicative inequality, for which the choice probability

vector for every menu is pivotal, in the sense defined in Section 2.2.

Third, by arranging that every participant saw only one menu from each choice domain, we obtain data on population choices that are plausibly independent as well as identically distributed, menu by menu.

5 The experimental dataset we have collected complements earlier work, including the abstract choice models introduced in McCausland & Marley (2013) and the inferential methods described in McCausland & Marley (2014). The data, supported by these models and methods, serve as a testing ground for the conditions we have considered in this paper and any others that one may wish to investigate. That said, the data are publicly available and stand alone, amenable to analysis
10 using other models and methods.

Evidence in support of regularity and random utility is fairly consistent across choice domains. While the evidence in favour of these conditions for any one choice domain is not strong, the direction of the evidence is robust to the choice of prior distribution and the collective evidence is quite favourable. While the literature on the attraction effect provides evidence against regularity
15 and therefore random utility in special cases, these conditions hold up across the wide variety of choice domains we explore, including a few domains designed to elicit the attraction effect. This is reassuring given the prevalence of random utility models in the literature.

Evidence regarding the multiplicative inequality, in contrast, varies considerably across domains. There is very strong evidence against it for three domains, so we can conclude that it is not
20 a promising candidate for a universal condition governing probabilistic discrete choice. Tversky (1972*b*) introduced the EBA class of models to accommodate the similarity effect, introduced in the same paper, but since the multiplicative inequality is a necessary condition for EBA models, there are other observed choice patterns that it cannot capture well. At the same time, the multiplicative inequality is favoured more than random utility in many domains, which raises the question of what
25 common features these domains might have. In future work we wish to investigate the multiplicative inequality in more detail, using a variety of data sources, complementing some theoretical work on that condition in McCausland & Marley (2024).

The dataset can be reused to evaluate other conditions on probabilistic choice behaviour, including behavioural axioms and conditions implied by different competing theories. A better un-
30 derstanding of choice behaviour requires evidence not only about specific anomalies, but also the broad applicability of such conditions. By providing a diverse population-level choice dataset, the present study aims to contribute to this endeavour.

	a_1	a_2	b	$E[\alpha]$	$\text{Var}[\alpha]$	σ_α	$E[\lambda]$	$\text{Var}[\lambda]$	σ_λ
M_1	2.00	1.00	3.00	9.00	27.00	5.196	0.667	0.0556	0.236
M_2	2.50	0.50	3.00	9.00	27.00	5.196	0.883	0.0347	0.186
M_3	2.75	0.25	3.00	9.00	27.00	5.196	0.917	0.0191	0.138

Table 4: For the three encompassing models M_1 , M_2 and M_3 , the table shows the values of three hyper-parameters, a_1 , a_2 and b , defining the prior distribution of α and λ ; the prior mean, variance and standard deviation of α ; and the same prior moments of λ .

A Prior distributions used in the prior sensitivity analysis

Table 4 gives the exact specification of the prior distributions of α and λ for the encompassing models M_1 , M_2 and M_3 used in the prior sensitivity analysis in Section 5.2. Each prior is defined in terms of the value of three tabulated hyper-parameters, a_1 , a_2 and b . The table also provides
5 the prior mean, variance and standard deviation of α and λ for each of the encompassing models.

In each case, α and λ are *a priori* independent. The prior distribution of the concentration parameter α is Gamma with shape parameter $a_1 + a_2$ and rate parameter b . The prior distribution of the dependence parameter λ is Beta with shape parameters a_1 and a_2 .

Moving from M_1 to M_3 increases prior concentration near $\lambda = 1$, while holding the prior
10 distribution of α constant. The priors were chosen to represent progressively stronger prior belief in dependence of choice probabilities across menus.

B Domains

We describe here the complete set of 32 choice domains featured in the experiment. We classify choice domains into four categories. In the first category, objects have no numerical attributes. In
15 the second category, objects have a single attribute, whose level is not explicitly given. In the third category, objects have two numerical attributes and the experimental design is intended to elicit one or more context effects. In the fourth category, objects have many attributes, and they are chosen to resemble objects in discrete choice experiments used in Marketing.

The section titles, “Male stars” for example, give the names of the domains for the purpose of
20 reporting results. The specification of a domain is shown in a grey box and consists of a question and five choice objects, or responses. In the actual experiment, for a given domain, all participants see the same question (such as “Which movie star would you choose to have lunch with?”) but not the same set of responses; different participants will see from two to five of the possible responses, in random order, as described in Section 3. For the purpose of reporting results, the choice objects in

each domain are labelled a , b , c , d and e , in the order presented in the grey boxes. The participants do not see these labels, and accordingly, we do not use them within the boxes.

We adopt the convention that whenever there is a natural order for the items in a list, we order them in decreasing order. Again, this is only for the purposes of reporting results; each participant
5 sees the items in a menu in random order.

B.1 No numerical attributes

B.1.1 Male stars

The source for this domain is the website **ranker.com**, accessed June 4, 2017. The list is “The best actors working today”. The choices are the top five actors in that list, in order.

Which movie star would you choose to have lunch with?

- Tom Hanks
- Kevin Spacey
- Morgan Freeman
- Leonardo DiCaprio
- Christian Bale

B.1.2 Female stars

The source for this domain is the website **ranker.com**, accessed June 4, 2017. The list is “The best American actresses working today”. These are the top five actors in that list, in order. Jodie Foster’s name was misspelled in the experiment, as two participants noted in the comments.

Which movie star would you choose to have lunch with?

- Meryl Streep
- Jody Foster
- Kathy Bates
- Amy Adams
- Julianne Moore

B.1.3 Films

The source for this domain is the IMDb list “Most Popular Feature Films Released 1990 to 1999”. The decade was chosen so that the films would not be easily recognizable by most respondents.

Judging from the following descriptions of films, which one of the films would you choose to see?

- Two imprisoned men bond over a number of years, finding solace and eventual redemption through acts of common decency.
- Mathilda, a 12-year-old girl, is reluctantly taken in by Léon, a professional assassin, after her family is murdered. Léon and Mathilda form an unusual relationship, as she becomes his protégé and learns the assassin's trade.
- The lives of two mob hit men, a boxer, a gangster's wife, and a pair of diner bandits intertwine in four tales of violence and redemption.
- A sexually frustrated suburban father has a mid-life crisis after becoming infatuated with his daughter's best friend.
- Identical twins, separated at birth and each raised by one of their biological parents, discover each other for the first time at summer camp and make a plan to bring their wayward parents back together.

B.1.4 Star pairs

Here, choice objects are pairs of movie stars from a set of four movie stars: Tom Hanks, Scarlett Johansson, Brad Pitt and Angelina Jolie. The only missing pair is Brad Pitt and Angelina Jolie.

- 5 One possible measure of similarity is the number of actors in common between two pairs, with values 0 and 1. There are nine doubleton choice sets (i.e. pairs of actor pairs) with one star in common and one ($\{c, d\}$) without any stars in common. Thus there are three triples ($\{a, c, d\}$, $\{b, c, d\}$, $\{c, d, e\}$) where one might expect a similarity effect.

- 10 In this example, respondents' preferences may depend not only on their liking of particular actors but also on complementarities between actors.

Knowing only who is starring, which one of these new films would you choose to see?

- Tom Hanks and Scarlett Johansson
- Scarlett Johansson and Brad Pitt
- Tom Hanks and Brad Pitt
- Scarlett Johansson and Angelina Jolie
- Tom Hanks and Angelina Jolie

B.1.5 Pizzas

The source for this domain is a Montreal pizza restaurant. All these pizzas are either 12 or 13 dollars.

Which one of the following pizzas would you choose?

- Mozzarella, tomato sauce, basil
- Pepperoni, mushrooms, green pepper, mozzarella, tomato sauce
- Red onion, tomato sauce, feta, mozzarella, olive oil, Greek spices, tomato sauce
- Bacon, white onion, mozzarella, parmesan, fresh cream, tomato sauce, ground pepper
- Mushrooms, green pepper, mozzarella, tomato sauce

B.1.6 Juices

Which one of the following fresh juices would you choose?

- Mango
- Orange
- Apple
- Grapefruit
- Pineapple

B.1.7 Colours

Which one of the following colours do you like best?

- Red
- Purple
- Pink
- Blue
- Green

5

B.1.8 Colour combos

The source for this domain is the website “The top tens”, page “Two colors that look good side by side.” The color combinations here are ranked 1, 4, 5, 13 and 14. We chose a selection of five high ranking combinations among which there were many colors in common. Using a similarity measure
10 equal to the number of colours in common between two pairs, there are two doubleton choice sets where the two colour combinations have no colours in common ($\{a, e\}$ and $\{b, d\}$) and eight where the two colour combinations have one colour in common. This gives six triplet pairs in which one might expect a similarity effect. The name for this domain used in data files and documents, “Colour combos” is an abbreviation of the title seen by participants, “Colour combinations”.

Which one of these colour combinations do you like best?

- Black and red
- Black and purple
- Black and blue
- Blue and red
- Blue and purple

B.1.9 Events

This domain involves comparisons of the probabilities of future events. Logically, the probability of event e must be at least as great as the probability of a, which must in turn be at least as great as the probability of d; also, the probability of b must be at least as great as the probability of c. There is potential for several asymmetric dominance effects in this domain, based on these dominance relations. These would be unconventional effects, as most experimental designs in the literature intended to elicit asymmetric dominance effects feature numerical dominance relations.

Which one of the following events do you think is most likely to happen in the next twenty years?

- Scotland becomes an independent country.
- Either Catalonia or Quebec become independent countries.
- Catalonia becomes an independent country.
- Scotland and Quebec become independent countries.
- Either Scotland or Quebec become independent countries.

B.1.10 Radio formats

The domain is radio formats, and the choice objects are the top 5 radio formats in Canada in 2015, according to

<https://byrnesmedia.com/2015/10/05/the-6-best-performing-radio-formats-in-canada/>

Descriptions of formats are from

<http://www.newsgeneration.com/broadcast-resources/guide-to-radio-station-formats/>

Suppose you were on a two hour road trip and you have a choice among radio stations with the following formats. Which one would you choose?

- News
- Hot Adult Contemporary, or Hot AC (A variety of classic and contemporary mainstream music geared towards adults.)
- Classic Hits (Rock and pop, roughly 1964-1989)
- Country Music
- Adult Contemporary, or AC (Adult-oriented pop/rock with no hard rock.)

B.1.11 Music

The choice objects in this domain are the top selling musical artists of all time, according to Wikipedia. They should be familiar to a large majority of respondents. The name for this domain
5 used in data files and documents, “Music” is an abbreviation of the title seen by participants, “Musical artists”.

Which one of the following musical artists do you like the best?

- The Beatles
- Elvis Presley
- Michael Jackson
- Madonna
- Elton John

B.1.12 Aboriginal

The name for this domain used in data files and documents, “Aboriginal” is an abbreviation of the
10 title seen by participants, “Aboriginal art”.

Which one of the following examples of Australian aboriginal art do you like the best?



B.1.13 Impressionist

The name for this domain used in data files and documents, “Impressionist” is an abbreviation of the title seen by participants, “Impressionist art”.

Which one of the following examples of Impressionist art do you like the best?



B.1.14 Sentences

This domain consists of sentences that have been used in experiments of linguistic judgement. They are, respectively sentences 7j, 7h, 7p, 7a and 7e in Bard et al. (1996). Figure 1 of that paper shows acceptability scores for these and other sentences given by two individual linguists, an acceptability score aggregating the scores of four linguists and an acceptability score aggregating the scores of four “naive respondents”, all undergraduate anatomy students. There is broad, but not perfect, agreement in terms of order, and in the following list, they are in decreasing order of acceptability according to the measure aggregating the judgements of four linguists.

Which one of the following sentences do you find the most grammatically acceptable?

- Who did Bill buy the car to please?
- This is a book which reading would be fun.
- Where did Bill buy the car to drive?
- Which man do you wonder when to meet?
- With which pen do you wonder what to write?

B.1.15 Travel

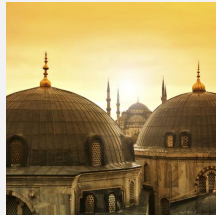
The source is Tripadvisor. These are the top five travel destinations, according to the results of an on-line contest where visitors to a Tripadvisor site could make pairwise choices between travel destinations.

Which one of the following travel destinations would you most like to visit?

1. Marrakech, Morocco



2. Istanbul, Turkey



3. Hanoi, Vietnam



4. Siem Reap, Cambodia



5. Prague, Czech Republic



B.1.16 Marijuana

This question elicits policy preferences.

Which one of the following marijuana policies would you choose?

- Possession by, and sales to adults are both legal; sales to minors are illegal.
- Possession by, and sales to adults are both illegal but neither is a criminal offense; sales to minors are a criminal offense.
- Possession is illegal but not criminal; all sales are a criminal offense.
- Possession and sales are criminal offenses, with a small number of medical exceptions.
- Possession and sales are criminal offenses, without exception.

B.2 Single attributes, not directly observed

The five domains in this section contain choice objects with an objective rank order.

B.2.1 Latitude

- 5 The five cities of this domain have a latitude close to 50 degrees north. In the following list, they are ordered from furthest north to furthest south. According to Wikipedia, their latitudes are, respectively, $52^{\circ}14'N$, $51^{\circ}30'N$, $49^{\circ}15'N$, $48^{\circ}51'N$ and $47^{\circ}36'N$. There are two potential asymmetric dominance effects, with Vancouver being fairly obviously north of Seattle and London being fairly obviously north of Paris.

Which one of the following cities do you think is furthest north?

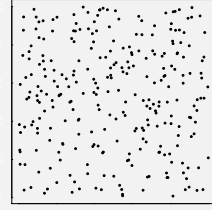
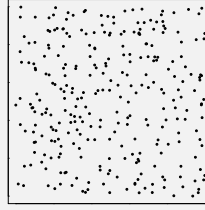
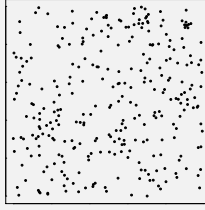
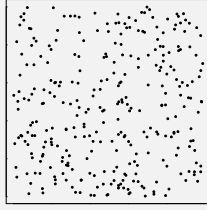
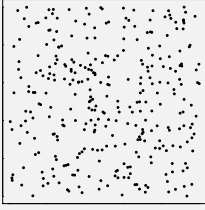
- Warsaw, Poland
- London, United Kingdom
- Vancouver, Canada
- Paris, France
- Seattle, United States

10

B.2.2 Dots

This domain is a perception example. The true numbers of points are, respectively, 320, 310, 300, 290 and 280. It is much clearer that there are more points in the first panel than in the fifth, than that there are more points in the first than in the second. The difference in the number of points
15 is an obvious similarity measure here that might be expected to lead to similarity effects.

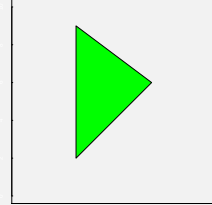
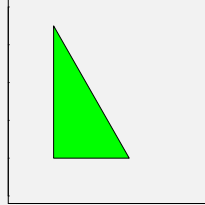
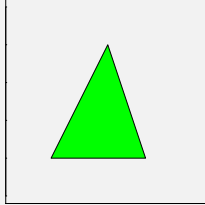
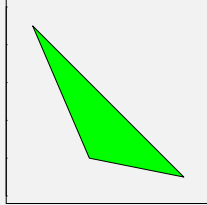
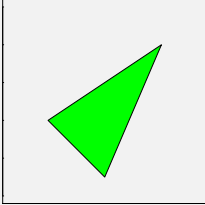
Which one of the following boxes do you think has the greatest number of points?



B.2.3 Triangles

This domain is another perception example. The true areas are, respectively, 16, 15, 15, 14 and 14 units.

Which one of the following triangles do you think has the greatest area?



5

B.2.4 Population

These countries were ranked, respectively, 4th through 8th in terms of population in 2016, when their populations, in millions, were 258, 206, 202, 186 and 156.

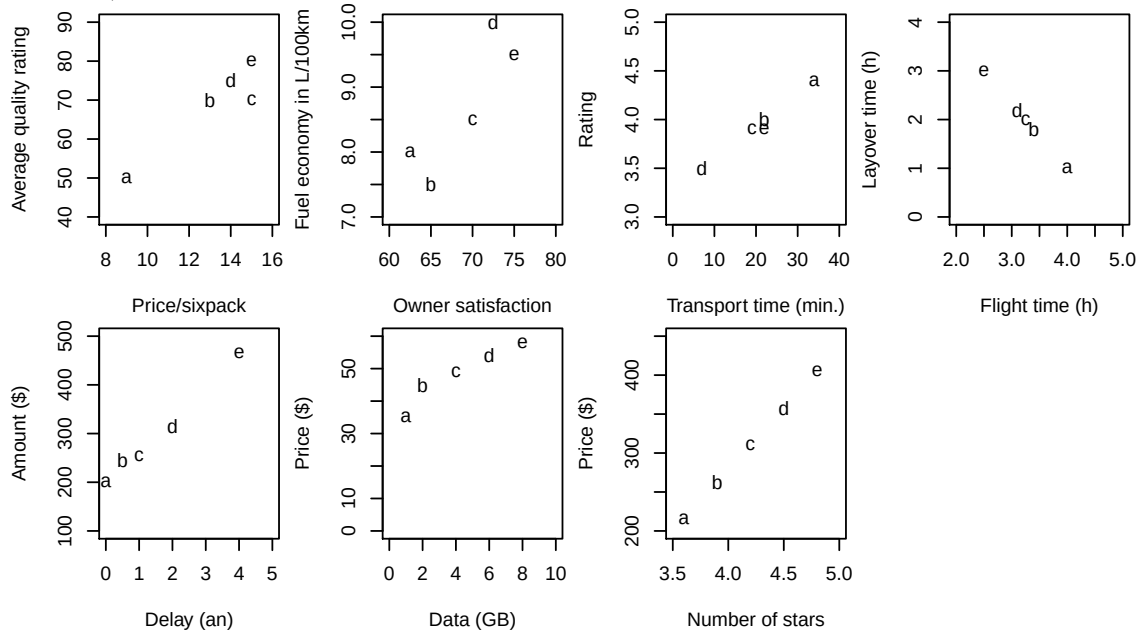
Which one of the following countries do you think has the largest population?

- Indonesia
- Brazil
- Pakistan
- Nigeria
- Bangladesh

10 B.2.5 Surface area

These countries are ranked, respectively, 2nd through 6th in terms of surface area, including inland bodies of water. In millions of square kilometres, those surface areas are 9.984, 9.573, 9.525, 8.516 and 7.692.

Figure 5: Designs for context effects. Each panel plots the five objects of a domain in attribute space. The domains are, in row major order, Beer, Cars, Restaurants, Flights I, Delayed Choice, Phone Plans, Hotels



Which one of the following countries do you think has the greatest surface area, including inland bodies of water?

- Canada
- United States of America
- China
- Brazil
- Australia

B.3 Objects with two attributes

Here we describe domains whose choice objects have exactly two real-valued attributes. In all cases, the levels of these attributes are presented directly to the participant. Figure 5 plots the five objects of each of these domains in attribute space, revealing potential conditions for context effects.

Table 5 shows, for the first four of these domains, dominance, similarity and betweenness relations that we might expect to generate context effects. Following usual practice, we exclude similarity relations for two objects in which there is also a dominance relation between the two; and betweenness relations among three objects when two of them are similar and the third is not similar to either of the first two.

Domain	Dominance	Similarity w/o dominance	Betweenness w/o similarity
Beer	$e > c, b > c, d > c$	$b \sim d, d \sim e, b \sim e$	$a b e$
Cars	$a > d, b > e$		$a c b, a c e, a c b, d c e$
Restaurants	$b > e, c > e$	$b \sim c$	$a b d, a c d, a e d$
Flights I		$b \sim c, c \sim d, b \sim d$	$a b e, a c e, a d e$

Table 5: Relations of dominance, similarity and betweenness in two-attribute domains

B.3.1 Beer

This domain is from an experiment reported in Huber et al. (1982) used to illustrate an asymmetric dominance effect. The prices are multiplied by 5 and we added choice objects d and e to allow for two more asymmetric dominance effects and two similarity effects. The first panel of Figure 5 show the choice objects in attribute space. Table 5 shows the relations of dominance, similarity and betweenness among objects associated with context effects. The most commonly used domains to illustrate the attraction effect involve Beer, Cars, Apartments, Computers, Restaurants and Televisions.

Below you will find three brands of beer. You know only the price per sixpack and the average quality ratings made by respondents in a blind taste test. Given that you had to choose one brand to buy on this information alone, which one would you choose?

Price/sixpack	Average quality rating (100 = Best; 0 = Worst)
\$9.00	50
\$13.00	70
\$15.00	70
\$14.00	75
\$15.00	80

B.3.2 Cars

This domain is based on an experiment from Wedell & Pettibone (1996). There are two experiments involving cars in that article, the experiment in question is numbered 18 in the appendix to that paper. Objects below have the same attributes as in that experiment and a similar range of levels. We adapted the objects to allow for compromise effects. The second panel of Figure 5 shows the choice objects in attribute space. Table 5 shows the relations of dominance, similarity and betweenness among objects associated with context effects.

Which one of the following cars would you choose to drive, all other features being equal?

Ride quality is on a scale of 0 to 100.

Ride quality	Miles per gallon
60	30
80	24
70	27
55	28
75	22

B.3.3 Restaurants

This domain is based on another experiment in Wedell & Pettibone (1996), numbered 19 in the appendix to that paper. The third panel of Figure 5 shows the choice objects in attribute space.

5 Table 5 shows the relations of dominance, similarity and betweenness among objects associated with context effects.

Which one of the following restaurants would you choose for your next restaurant meal, based on transportation time (in minutes) and average customer ratings (from 1 to 5).

Transportation time	Rating
34	4.4
22	4.0
19	3.9
7	3.5
22	3.9

B.3.4 Layovers

The fourth panel of Figure 5 shows the choice objects in attribute space. Table 5 shows the relations
10 of dominance, similarity and betweenness among objects associated with context effects. The name for this domain used in data files and documents, “Layovers” is an abbreviation of the title seen by participants, “Flight layovers”.

Which one of the following flight itineraries would you choose? All involve two flights, with one layover between them.

Total inflight time	Layover time	Total itinerary time
4:00	1:00	5:00
3:24	1:48	5:12
3:15	2:00	5:15
3:06	2:12	5:18
2:30	3:00	5:30

B.3.5 Future

This domain is loosely based on an experiment by Benzion et al. (1989), in which respondents are asked to assign present values equivalent to the receipt of \$200 at time horizons of 0.5, 1, 2 and 4 years. Based on implied discount factors at various terms, we constructed five choice objects designed to have approximately the same present value equivalent. The name for this domain used in data files and documents, “Future” is an abbreviation of the title seen by participants, “Future payments”.

Which one of the following would you choose?

- \$200 credited to your bank account immediately.
- \$245 credited to your bank account in six months.
- \$255 credited to your bank account in one year.
- \$315 credited to your bank account in two years.
- \$465 credited to your bank account in four years.

B.3.6 Phone plans

The source for this domain is the website of Fido Mobile, a Canadian cell phone carrier, with rates quoted on March 1, 2017 in Canadian dollars.

Of the following cell phone plans, which one would you choose? In all cases, unlimited calling, text picture and video messages to Canadian and international mobile numbers are included. Excess data usage is billed at \$10 per 500 MB.

- 1 GB data per month, \$35 per month.
- 2 GB data per month, \$45 per month.
- 4 GB data per month, \$49 per month.
- 6 GB data per month, \$54 per month.
- 8 GB data per month, \$58 per month.

B.3.7 Hotel rooms

Using Expedia results, I did a linear regression of price per night on a constant and the Expedia rating, in numbers of stars, for a sample of available hotels. The five levels of numbers of stars
5 correspond roughly to the mean, plus and minus one sample standard deviation and plus and minus two standard deviations. Prices are approximately equal to fitted values in the linear regression.

Suppose you are staying over two nights in New York city. Which one of the following hotels would you choose, based on customer ratings and price per night?

- 3.6/5 stars, \$215 per night
- 3.9/5 stars, \$263 per night
- 4.2/5 stars, \$311 per night
- 4.5/5 stars, \$358 per night
- 4.8/5 stars, \$406 per night

B.4 Objects with multiple attributes

B.4.1 Itineraries

- 10 This domain illustrates three-way tradeoffs. The points form a constellation in the simplex that resembles the pattern of points on the “five” side of a die. The name for this domain used in data files and documents, “Itineraries” is an abbreviation of the title seen by participants, “Two-flight itineraries”.

Which one of the following flight itineraries would you choose? All involve two flights and have a total duration of six hours.

1st flight	Layover	2nd flight
1:30	1:15	3:15
3:15	1:15	1:30
2:15	1:30	2:15
1:30	1:45	2:45
2:45	1:45	1:30

B.4.2 Televisions

The source for this domain is the website of Best Buy Canada, with prices in Canadian dollars.

Which one of the following televisions would you choose to buy if you were in the market for a television? All are LED televisions. Resolution refers to number of horizontal lines. Smart indicates internet connectivity.

Brand	Resolution	Smart	Price (\$)	Screen Size (inches)
Sharp	1080	Yes	309	32
Insignia	720	No	209	32
Sony	720	Yes	439	32
Samsung	1080	Yes	459	40
Toshiba	1080	No	409	43

5 B.4.3 Coffee

The source for this domain is the website `buycoffeecanada.com`, with prices in Canadian dollars.

You need to buy 16oz of ground coffee for a brunch with friends. Which one of the following ground coffees would you choose?

Price	Fair Trade	Name: Description
18.71	Yes	Ethiopian Yirgacheffe: vibrant and intensely aromatic, fruity
9.99	No	Colombian Supremo: mellow cup, complex aromas and rich flavours
13.72	Yes	Colombian Organic: medium body, fragrant aroma and mild acidity
12.35	No	Tanzania Peaberry: full bodied coffee, chocolatey aroma, wine-like finish
13.46	No	Sumatra Mandheling: exotic, earthy, bright with low acidity

B.4.4 Charity

The charities in this domain are real charities. The first two have broadly supported health related missions. However, the Canadian Cancer Society attracts much more financial support than Arthritis Research Canada and we might expect that most people would prefer a marginal dollar going to the former. The two other charities have goals that are nearly opposite and elicit strong emotions (of different kinds) from some.

Suppose someone was donating a total of 100 dollars to a combination of charities, on your behalf. Which one of the following divisions of the 100 dollars would you choose?

Arthritis Research Canada	Canadian Cancer Society	Canadian Coalition for Firearm Rights	Coalition for Gun Control
90	10	0	0
35	60	5	0
35	60	0	5
10	80	10	0
10	80	0	10

Statements and Declarations

Supplementary Materials

Supplementary Materials can be found in the folder `Supplementary_materials` of the GitHub repository <https://github.com/mccauslw/Population>.

Figures analogous to Figure 4 and tables analogous to Table 1 for each of the 32 domains are in the document `all_domains_data.pdf`. A copy of the consent form that participants must complete before participating in the experiment is in the file `Consent_form.pdf`. A copy of the Certificate of Approval issued by Human Research Ethics Board at the University of Victoria is in the file
5 `Certificate_of_approval.pdf`. A document describing in more detail the recruitment and compensation procedures followed by SSI is in the file `Recruitment_compensation_procedures.pdf`.

Authors' Contribution Statement

McCausland contributed to conceptualization, data curation, formal analysis, funding acquisition, investigation, project administration, software, visualization and writing (initial draft, review and
10 editing). Marley contributed to the conceptualization, funding acquisition, investigation, project administration, and writing-review and editing. Davis-Stober contributed to the conceptualization, investigation, and writing - review and editing.

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Competing Interests

The authors declare none.

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